

FINITE CONGRUENCE COVERINGS FOR THE ERDŐS–STRAUS EQUATION AND STERILE PROFINITE POINTS

ANONYMOUS

ABSTRACT. The Elsholtz–Tao families of residue classes on which the Erdős–Straus equation $4/n = 1/x + 1/y + 1/z$ is solvable by polynomials are clopen subsets of $\widehat{\mathbb{Z}}$. By compactness and Dirichlet’s theorem, all but finitely many primes of a given set admit a finite covering by polynomial identities exactly when its set of accumulation points in $\widehat{\mathbb{Z}}^\times$ contains no *sterile* point, i.e. no point outside every class. The only sterile points known to us are the square points, which give the classical Mordell–Schinzel obstruction. We collect what is known about coverings of sets of primes that avoid the square points by a non-residue condition at one further prime r . For Type I coverings of the hard primes with least non-residue r , a covering exists for $r = 5$ (of height 10), and the least covering height equals the limsup of the Type I depth under Hypothesis H. For $r = 7$ every covering has height greater than $1.32 \cdot 10^{12}$ (finite computation), because of an explicit “sign point” at which the two lowest admissible 2-adic levels are proved empty. For $r = 13$ we give an explicit finite-exception theorem, a repackaging of Salez’s level sieve. We show that a point that lies in no class of modulus at most 10^6 , and was earlier conjectured to be sterile, does lie in an explicit class of modulus 12670944. A new candidate survives targeted searches. For $r = 17$ the classes meeting the 17-generic line are balls around solutions of $4/17^K = 1/x + 1/y + 1/z$; an explicit bound on the number of such solutions would produce a sterile point and rule out every finite polynomial covering of the Mordell-hard primes with least non-residue 17. The bound needed is a discrete-logarithm counting problem, which we leave open. Nothing here proves or disproves the Erdős–Straus conjecture.

1. INTRODUCTION

The Erdős–Straus conjecture (ES) asserts that for every integer $n \geq 2$

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \tag{1}$$

has a solution in positive integers. It is open; it suffices to treat primes. The oldest approach is by *polynomial identities*: on a residue class one writes down x, y, z as polynomials in n . Mordell’s identities settle every prime outside the six square classes 1, 121, 169, 289, 361, 529 modulo 840 [8], and Mordell and Schinzel showed that a square class can never be settled

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this way [8, 22]. Elsholtz and Tao essentially classified the classes that are solvable by polynomials, by seven families [4, Prop. 1.9], and remarked that the odd-square vanishing “rules out . . . a finite set of covering congruence strategies” [4, p. 6]. Salez [21] and, later, Mihnea and Dumitru [7] sieve with these classes to great depth.

This note looks at finite coverings from the profinite side. Every ET class is a clopen subset of $\widehat{\mathbb{Z}}$, so for a closed target set compactness reduces the existence of a finite covering to a pointwise question: is every point in some class? A unit point that lies in no ET class is called *sterile*. Square points (squares of units at every prime) are sterile; this is the Mordell–Schinzel obstruction. Once the target excludes them, for instance by asking p to be a non-residue modulo one further prime r , the question is whether *other* sterile points exist. We consolidate the project’s internally reviewed work on this question.

Results.

- *Framework* (§2, PROVED). For a set $\mathcal{P}$ of primes with accumulation set $\mathcal{P}' \subset \widehat{\mathbb{Z}}^\times$, a finite covering of all but finitely many $p \in \mathcal{P}$ by primitive classes solvable by polynomials exists if and only if $\mathcal{P}'$ has no sterile point (Proposition 2.5). For a single explicit point only one direction holds (Remark 2.6).
- *Square points* (§3). Square points are sterile (PROVED; classical). The project’s procedure-level generalisation, Theorem C of [12], is stated with its H-conditional label and hedged novelty.
- *Type I and the least non-residue* (§4). For the Type I depth $ck_{\min}(p)$ of hard primes with least non-residue r : $C(5) = 10$ (PROVED); under Hypothesis H the limsup $C^*(r)$ equals the least height of a finite Type I covering, and it is infinite if and only if a Type I-sterile point exists (Theorem 4.2). At the sign point $\hat{x}_9$ no certificate has its smaller divisor below 10^{12} (CERTIFIED), so every Type I covering for $r = 7$ has height $> 1.32 \cdot 10^{12}$, and $C(7) > 1.32 \cdot 10^{12}$ under H. The 2-adic levels 5 and 6 are empty (PROVED). Certificates in the sign fibre are norm-one units of a Pell order of a special shape (PROVED), and none at $\hat{x}_9$ has 2-adic level ≤ 22 and $v_7(k) \leq 3$, at any height (CERTIFIED). At the levels 7–10 a certificate at $\hat{x}_9$ needs $v_7(k) \geq 8$ and lies in one explicit two-parameter regime (PROVED and CERTIFIED). The Type I-sterile set is nowhere dense (PROVED), and a measure criterion would give $C(7) = \infty$ under H (PROVED reduction; hypothesis open).
- $r = 13$ (§5). Primes with $(p/13) = -1$ satisfy ES outside six classes modulo 720720 (PROVED by certified finite computation; an explicit packaging of Salez’s sieve). The point x^* with $x_{11}^* = x_{13}^* = 2$ lies in no ET class of modulus $\leq 10^6$ (CERTIFIED). It is nevertheless not sterile: it lies in an explicit II3 class of modulus 12670944 (Proposition 5.4, PROVED). This refutes an earlier conjecture. The point x^{**} with

$(x_{11}^{**}, x_{13}^{**}) = (2, 15)$ survives targeted searches far beyond that height (CERTIFIED within stated ranges); that it is sterile is a conjecture supported only by this evidence.

- $r = 17$ (§6). On the 17-generic line the classes become balls around solutions of $4/17^K = 1/x + 1/y + 1/z$ (PROVED); levels ≤ 5 leave 67.7% of each non-residue cell uncovered (CERTIFIED); an explicit tail bound would give a sterile point and rule out every finite polynomial covering of the Mordell-hard primes with $n_p = 17$ (Theorem 6.4, PROVED reduction; Corollary 6.5, CONDITIONAL). The missing bound is a discrete-logarithm count (ASSESSMENT).

Section 7 lists open problems. No sterile point other than a square point is known to us. The literature expects the opposite at the level of integers: Yamamoto [25, p. 47] writes that “it is believed that any positive integer, which is a quadratic residue of 3, 5, 7 and 8, and still is not a perfect square, is contained in some covering”, and Ionascu and Wilson [5] conjecture that the excluded residues of ever larger moduli are eventually all squares or composite. These are statements about integers and residues. A sterile non-square point of $\widehat{\mathbb{Z}}^\times$ would not contradict them (O101 audit). A sterile point would show that a particular set of primes cannot be handled by finitely many polynomial identities; it would say nothing about whether ES holds for those primes.

Labels. PROVED means proved here or in the cited source, with the proof checked internally. CERTIFIED means established by a finite computation; we name the engines and say whether the computation was reproduced independently. CONDITIONAL results assume a named hypothesis (Schinzel’s Hypothesis H [23], or the hypothesis of a stated theorem). CONJECTURE, EVIDENCE and ASSESSMENT (a heuristic judgement) are not results.

2. COVERINGS AS CLOPEN CLASSES

2.1. The seven families. We use the families of Elsholtz and Tao [4, Prop. 1.9]. Each is indexed by a triple κ of positive integers subject to a side condition, and defines a set of residue classes:

family	parameters	side condition	class
I1	a, d, f	$f \mid 4a^2d + 1$	$n \equiv -f \pmod{4ad}$
I2	a, c, f	$(4ac, f) = 1$	$n \equiv -f \pmod{4ac}, n \equiv -c/a \pmod{f}$
I3	c, d, f	$(4cd, f) = 1$	$n \equiv -f \pmod{4cd}, n^2 \equiv -4c^2d \pmod{f}$
I4	a, b, e	$e \mid a + b, (e, 4ab) = 1$	$n \equiv -1/e \pmod{4ab}$
II1	a, b, e	$e \mid a + b, (e, 4ab) = 1$	$n \equiv -e \pmod{4ab}$
II2	a, d, f	$4ad \mid f + 1$	$n \equiv -4a^2d \pmod{f}$
II3	a, d, e	$(4ad, e) = 1$	$n \equiv -4a^2d - e \pmod{4ade}$

On its class each family carries explicit solutions. With $\pi^I(a, b, c, d) = (abdn, acd, bcd)$ and $\pi^{II}(a, b, c, d) = (abd, acdn, bcdn)$ [4, §2], the remaining coordinates are

I1	$e = (4a^2d + 1)/f, b = (ne + 1)/(4ad), c = (n + f)/(4ad)$
I2	$d = (n + f)/(4ac), b = (na + c)/f$
I3	$a = (n + f)/(4cd), b = (n + (n^2 + 4c^2d)/f)/(4cd)$
I4	$c = (a + b)/e, d = (ne + 1)/(4ab)$
II1	$c = (a + b)/e, d = (n + e)/(4ab)$
II2	$c = (f + 1)/(4ad), b = (nc + a)/f$
II3	$c = (n + 4a^2d + e)/(4ade), b = ce - a = (n + e)/(4ad)$

and the solution is π^I for I1–I4 and π^{II} for II1–II3.

Lemma 2.1 (PROVED; [4, §§2, 10]). *Let κ be an admissible parameter triple of one of the seven families and let $n \geq 1$ be an integer in its class. Then the coordinates above are positive integers, and the resulting (x, y, z) is a solution of $4/n = 1/x + 1/y + 1/z$ in positive integers.*

Proof. The identity $4/n = 1/x + 1/y + 1/z$ follows from the defining relations of the varieties $\Sigma_n^I, \Sigma_n^{II}$ of [4, (2.1)–(2.9), (2.13)–(2.21)]. Integrality on the class is a direct check; for example in I3, $n \equiv -f \pmod{4cd}$ gives $a \in \mathbb{Z}$, the second congruence gives $Y := (n^2 + 4c^2d)/f \in \mathbb{Z}$, and $fY \equiv f^2 \pmod{4cd}$ with f invertible modulo $4cd$ gives $Y \equiv f \equiv -n$, so $b \in \mathbb{Z}$. With κ fixed, every coordinate is a polynomial in n with non-negative coefficients that is positive at $n = 1$, so all coordinates are positive for $n \geq 1$. $\square$

The positivity statement for all $n \geq 1$ (not only for large n) was checked in the project’s review of [18]; the degrees of (x, y, z) in n are at most 4.

ET call a primitive residue class *solvable by polynomials* if there are polynomials P_1, P_2, P_3 with rational coefficients, taking positive integer values at all sufficiently large n of the class, with $4/n = \sum 1/P_i(n)$ identically. Lemma 2.1 says that every primitive class contained in one of the seven families is solvable by polynomials. We use the converse in the following form.

Cited theorem 2.2 ([4, Prop. 1.9]). *If a primitive residue class is solvable by polynomials, then all sufficiently large primes in it lie in the union of finitely many classes of the seven families.*

(ET state this separately for Type I and Type II solvable classes, with the families I1–I4 and II1–II3 respectively; every solvable primitive class is of one of the two types [4, p. 8].)

2.2. Profinite language. Let $\widehat{\mathbb{Z}} = \prod_q \mathbb{Z}_q$ be the profinite completion of $\mathbb{Z}$, with unit group $\widehat{\mathbb{Z}}^\times = \prod_q \mathbb{Z}_q^\times$. Every class of the table is defined by congruences modulo an integer $M(\kappa)$ (for I3, modulo $4cdf$); we write $\text{Cl}(\kappa) \subset \widehat{\mathbb{Z}}$ for the set of $x \in \widehat{\mathbb{Z}}$ satisfying them. It is open and closed (*clopen*), and an integer n lies in $\text{Cl}(\kappa)$ exactly when it lies in the class. We call $\text{Cl}(\kappa)$ an *ET class*.

Definition 2.3. A point $x \in \widehat{\mathbb{Z}}^\times$ is *sterile* if it lies in no ET class. For a subfamily $\mathcal{K}$ of parameter triples, x is $\mathcal{K}$ -*sterile* if it lies in no $\text{Cl}(\kappa)$, $\kappa \in \mathcal{K}$.

Let $\mathcal{P}$ be an infinite set of primes and $\mathcal{P}'$ its set of accumulation points in $\widehat{\mathbb{Z}}$.

Lemma 2.4 (PROVED). *$\mathcal{P}'$ is a compact subset of $\widehat{\mathbb{Z}}^\times$. If $\Sigma = \{x \in \widehat{\mathbb{Z}}^\times : x \bmod M \in R\}$ for a set R of reduced residues modulo M , and $\mathcal{P}(\Sigma)$ is the set of primes $p \nmid M$ with $p \bmod M \in R$, then $\mathcal{P}(\Sigma)' = \Sigma$.*

Proof. If $x \in \mathcal{P}'$, then for each prime q and each k infinitely many $p \in \mathcal{P}$ satisfy $p \equiv x \pmod{q^k}$; all but one of them are prime to q , so $x_q \in \mathbb{Z}_q^\times$. Derived sets in the compact metric space $\widehat{\mathbb{Z}}$ are closed, hence compact. For the second statement, a point of $\mathcal{P}(\Sigma)'$ is congruent modulo M to some element of $\mathcal{P}(\Sigma)$ and, by the first statement, is a unit, so it lies in Σ . Conversely every neighbourhood $x + Q\widehat{\mathbb{Z}}$ ($M \mid Q$) of $x \in \Sigma$ is a reduced class modulo Q and contains infinitely many primes by Dirichlet's theorem; they lie in $\mathcal{P}(\Sigma)$. $\square$

Such a Σ is clopen in $\widehat{\mathbb{Z}}^\times$, and by compactness every clopen subset of $\widehat{\mathbb{Z}}^\times$ has this form for some M . Note that $\widehat{\mathbb{Z}}^\times$ is closed but not open in $\widehat{\mathbb{Z}}$, so “clopen” always refers to the topology of $\widehat{\mathbb{Z}}^\times$. The sets of primes studied below are all of the form $\mathcal{P}(\Sigma)$. The prime p itself is never a unit of $\mathbb{Z}_p$, which is why we work with accumulation points rather than with the primes as points of $\widehat{\mathbb{Z}}$.

Proposition 2.5 (PROVED). *Let $\mathcal{P}$ be an infinite set of primes.*

- (a) *If no point of $\mathcal{P}'$ is sterile, then finitely many ET classes contain all but finitely many $p \in \mathcal{P}$. For those p , Lemma 2.1 gives an explicit solution of $4/p = 1/x + 1/y + 1/z$.*
- (b) *If some $x \in \mathcal{P}'$ is sterile, then no finite set of ET classes contains all but finitely many $p \in \mathcal{P}$. Moreover no finite set of primitive residue classes, each solvable by polynomials, contains all but finitely many $p \in \mathcal{P}$.*

Hence all but finitely many elements of $\mathcal{P}$ lie in finitely many primitive classes solvable by polynomials if and only if $\mathcal{P}'$ contains no sterile point.

Proof. (a) The ET classes are open and cover the compact set $\mathcal{P}'$, so finitely many of them cover it; let U be their union. If $\mathcal{P} \setminus U$ were infinite, it would have an accumulation point, which lies in $\mathcal{P}' \subset U$; but U is open and misses $\mathcal{P} \setminus U$, a contradiction.

(b) Removing finitely many elements from $\mathcal{P}$ does not change $\mathcal{P}'$. If finitely many ET classes contain a cofinite subset of $\mathcal{P}$, their union is closed and therefore contains x , which contradicts sterility. Next let $C_1, \dots, C_s$ be primitive classes solvable by polynomials whose union contains a cofinite subset of $\mathcal{P}$. Viewed in $\widehat{\mathbb{Z}}$, their union is closed, so $x \in C_j$ for some j . By Cited Theorem 2.2 there are ET classes $K_1, \dots, K_t$ containing all large primes of C_j . Their union is closed and misses x , so some neighbourhood $V = x + Q\widehat{\mathbb{Z}} \subset C_j$ misses it. As x is a unit, V is a reduced class modulo Q ,

and by Dirichlet's theorem it contains infinitely many primes. These lie in C_j and in no K_i , a contradiction. The final equivalence combines (a), (b) and Lemma 2.1: each ET class is a finite union of residue classes modulo $M(\kappa)$; the non-primitive ones contain at most finitely many primes, and each primitive one is solvable by polynomials by Lemma 2.1. $\square$

Remark 2.6 (Direction of the implications). The equivalence concerns the *existence* of a sterile point of $\mathcal{P}'$. For a *given* point x only one direction holds: if x is sterile there is no finite covering, but if x lies in some ET class another point of $\mathcal{P}'$ may still be sterile. An earlier internal draft of [18] said “equivalently” here; this was false and was corrected in review. A weaker statement is available for every explicit point: if $x \in \mathcal{P}'$ lies in no ET class of modulus at most Y , then every finite ET covering of a cofinite subset of $\mathcal{P}$ uses a class of modulus greater than Y . Indeed the classes of modulus $\leq Y$ in the covering miss a neighbourhood of x , which contains infinitely many elements of $\mathcal{P}$.

Remark 2.7 (Heights). The same compactness argument works for any subfamily $\mathcal{K}$ and any function $\kappa \mapsto \text{ht}(\kappa)$. Let $\mathcal{K}_X$ be the triples of height at most X and $U_X = \Sigma \setminus \bigcup_{\kappa \in \mathcal{K}_X} \text{Cl}(\kappa)$ for a compact Σ . The U_X are closed and decreasing. If each is nonempty, the finite intersection property gives a point of $\bigcap_X U_X$, which is $\mathcal{K}$ -sterile. So either some finite subfamily of bounded height covers Σ , or Σ has a $\mathcal{K}$ -sterile point. The set of $\mathcal{K}$ -sterile points of Σ is $\bigcap_X U_X$, a closed set.

3. SQUARE POINTS AND THE ODD-SQUARE OBSTRUCTION

Proposition 3.1 (PROVED; known). *Let $x \in \widehat{\mathbb{Z}}^\times$ be a square point: $x_q \in (\mathbb{Z}_q^\times)^2$ for every prime q . Then x is sterile.*

Proof. For $x = 1$ this is elementary: if $1 \in \text{Cl}(\kappa)$ then the integer $n = 1$ lies in the class, and Lemma 2.1 gives positive integers with $4 = 1/x + 1/y + 1/z \leq 3$. In general, suppose $x \in \text{Cl}(\kappa)$ with $\text{Cl}(\kappa)$ defined modulo M . For each $q \mid M$ the component x_q is the square of a unit, so by the Chinese remainder theorem there is an integer m prime to M with $m^2 \equiv x \pmod{M}$. The primitive class $m^2 \bmod M$ is then contained in the class of κ and is solvable by polynomials (Lemma 2.1). This contradicts the theorem of Mordell and Schinzel that a primitive class whose residue is a perfect square is not solvable by polynomials [8, 22], [4, p. 8]. $\square$

The last step is the classical obstruction; ET remark that it also follows from their vanishing result at odd squares [4, Prop. 1.6], which goes back to Schinzel and Yamamoto [25]. The input is one quadratic character computation; Bright and Loughran [1] identify it with the Brauer–Manin obstruction of the Erdős–Straus surfaces.

Call a prime p *Mordell-hard* if $p \bmod 840$ is a square, $p \bmod 840 \in \{1, 121, 169, 289, 361, 529\}$. Every primitive class modulo 840 that is not a square is solvable by polynomials (Mordell; see [4, p. 8]). The set $\Sigma_{\text{hard}} \subset \widehat{\mathbb{Z}}^\times$

of square classes modulo 840 contains the point 1, so by Propositions 2.5(b) and 3.1 the Mordell-hard primes admit no finite covering by polynomial identities. Only the elementary case $x = 1$ of Proposition 3.1 is used here, so this statement depends on Cited Theorem 2.2 but not on the Mordell–Schinzel theorem. This is the classical form of the obstruction, and it is the reason why every finite covering statement below restricts the hard primes further by a *non-residue* condition at some additional prime r ; the resulting target sets contain no square point. Whether they contain other sterile points is the question of this note.

3.1. The procedure-level form. The project’s working note [12] proves a version of the odd-square obstruction for procedures. We only state it informally and refer to [12, §§1–2] and [11] for the model of computation. A *witness-producing program* takes a prime p , performs ring operations, floor divisions, eventual-sign comparisons, full factorisations and loops over the lists so produced, and outputs a solution or FAIL; it is *correct* if every solution it outputs is genuine, and *bounded* if it uses only these operations, each loop running over a list fixed at loop entry (so it halts on every input; the number of steps may grow with p). Let $\mathcal{H}$ be the set of primitive irreducible $h \in \mathbb{Z}[X]$ with positive leading coefficient (denoted $\mathcal{P}$ in [12]). For $P \in \mathbb{Z}[X]$ a point $q^* \in \hat{\mathbb{Z}}$ is *square-mimicking* for P if $P(q_\ell^*)$ is the square of a unit of $\mathbb{Z}_\ell$ for every prime ℓ .

Cited theorem 3.2 (Formal odd-square principle [12, Theorem C, Corollary C1]; CONDITIONAL on Hypothesis H). Let $P = uX + v \in \mathcal{H}$ be linear and q^* square-mimicking for P and nondegenerate for every $h \in \mathcal{H}$ in the sense of [12, §1.2]. If a correct witness-producing program has a defined and finite formal run at (P, q^*) , then it outputs FAIL at every admissible q (PROVED); under Schinzel’s Hypothesis H [23] for an explicit finite family there are infinitely many admissible q , and $p = P(q) \equiv 1 \pmod{24}$. In particular, under H for one explicit finite family, every correct bounded witness-producing program fails for infinitely many primes $p \equiv 1 \pmod{24}$.

A finite list of ET classes, each tested by a congruence and followed by the identity of Lemma 2.1, is such a bounded program, and its failure at the rational square-mimicking point 1 is Proposition 3.1 for $x = 1$, unconditionally. Cited Theorem 3.2 does not reprove the unconditional statement for general square points; it extends the obstruction from congruence coverings to all bounded procedures, at the price of Hypothesis H. *Novelty.* Theorem C is the procedure-level, H-conditional generalisation of known obstructions (Mordell and Schinzel on square classes, ET Prop. 1.6 and the remark on [4, p. 6] that “a finite set of covering congruence strategies” must fail, Yamamoto, Bright–Loughran); its unconditional instances are known in substance. The project’s novelty audit rates it as a partial novelty only.

Remark 3.3 (ASSESSMENT). The target sets of §§4–6 require x_r to be a non-square for some prime r , so they contain no point that is square-mimicking

for $P = X$, and neither Proposition 3.1 nor Cited Theorem 3.2 applies to them. The candidate sterile points below (the sign point $\hat{x}_9$, the point x^{**} for $r = 13$, the 17-generic line) are therefore not explained by the odd-square obstruction. If they are sterile, the reason is a different, rigidity type of obstruction (§4); this is a heuristic reading, not a theorem.

4. TYPE I COVERINGS AND THE LEAST NON-RESIDUE

4.1. Slices, certificates and the depth $ck_{\min}$. In this section a prime p is *hard* if $p \equiv 1 \pmod{24}$, and n_p denotes the least quadratic non-residue modulo p (a prime ≥ 5). For an odd prime p let

$$\mathcal{B}_p = \{(c, k) : 1 \leq k \leq \lfloor 2p/3 \rfloor, 1 \leq c \leq \lfloor (2p+k)/4k \rfloor, (p, ck) = 1\},$$

put $N_{c,k} = p^2 + 4ck^2$, and let $M_{c,k}(p)$ be the number of divisors D of $N_{c,k}$ with $D \equiv -p \pmod{4ck}$. The Type I solutions of $4/p = 1/x + 1/y + 1/z$ (those with p dividing exactly one denominator) correspond to the pairs $((c, k), D)$ of this kind [10, §§26, 36]. In ET's coordinates a Type I solution has $(c, d, f)_{\text{ET}} = (k, c, D)$, and fixing (c, k, D) gives exactly a class of family I3. Write $s = \text{sf}(c)$ for the squarefree part of c and χ_s for the Kronecker character of $\mathbb{Q}(\sqrt{-s})$. If $\chi_s(p) = 1$ then $M_{c,k}(p) = 0$ [10, Thm. 48.1] (genus theory; the slice is *forced*). For hard p this applies to $s \in \{1, 2, 3, 6\}$, and to every c all of whose prime factors are residues modulo p . Put

$$ck_{\min}(p) = \min\{ck : (c, k) \in \mathcal{B}_p, M_{c,k}(p) > 0\},$$

the least height of a Type I solution in these coordinates. By the forcing just described, $ck_{\min}(p) \geq n_p$ for hard p . For a prime $r \geq 5$ put

$$C(r) = \sup\{ck_{\min}(p) : p \text{ hard}, n_p = r\}, \quad C^*(r) = \limsup_{p \text{ hard}, n_p=r} ck_{\min}(p).$$

A *certificate* is a triple $\kappa = (c, k, F)$ of positive integers with $(F, 4ck) = 1$ and $\text{sf}(c) \notin \{1, 2, 3, 6\}$; its *height* is ck , and its class is the ET class (family I3)

$$\text{Cl}(c, k, F) = \{x \in \widehat{\mathbb{Z}} : x \equiv -F \pmod{4ck}, x^2 \equiv -4ck^2 \pmod{F}\}.$$

If $(c, k) \in \mathcal{B}_p$, then $M_{c,k}(p) \geq 1$ if and only if $p \in \text{Cl}(c, k, F)$ for some F , and the possible F are exactly the divisors counted by $M_{c,k}(p)$ [14, Lemma 0.1]. Let $\Sigma_r \subset \widehat{\mathbb{Z}}^\times$ be the clopen set of x with $x \equiv 1 \pmod{24}$, $x \pmod{\ell}$ a non-zero square for the primes $5 \leq \ell < r$, and $x \pmod{r}$ a non-square. By quadratic reciprocity a prime $p > r$ is hard with $n_p = r$ if and only if $p \pmod{24 \prod_{\ell \leq r} \ell}$ lies in Σ_r ; by Lemma 2.4 these primes accumulate exactly on Σ_r . A *Type I covering of height $\leq X$* is a finite set of certificates of height $\leq X$ whose classes cover Σ_r ; let $X_r \in \mathbb{N} \cup \{\infty\}$ be the least height of a Type I covering.

The other Type I families I1, I2, I4 also give Type I solutions, but in each of them c or k grows linearly with p [13, §6.1]; only family I3 with fixed (c, k) bounds $ck_{\min}$.

4.2. The case $r = 5$.

Theorem 4.1 (PROVED; [13, Thm. 6.1]). *Every hard prime p with $n_p = 5$ has $ck_{\min}(p) \leq 10$, and $ck_{\min}(p) = 5$ if $p \equiv 2 \pmod{5}$. The bound is attained at $p = 193$, so $C(5) = 10$.*

Proof. Here $p \equiv 1 \pmod{8}$, $p \equiv 1 \pmod{3}$ and $p \equiv 2, 3 \pmod{5}$, so $p \equiv 17$ or $33 \pmod{40}$; all slices below lie in $\mathcal{B}_p$ as $p \geq 73$. If $p \equiv 17 \pmod{40}$, then $-p \equiv 3 \pmod{20}$ and $p^2 + 20 \equiv 0 \pmod{3}$, so $D = 3$ is a target divisor of the slice $(5, 1)$ and $ck_{\min}(p) \leq 5$. If $p \equiv 33 \pmod{40}$, then $-p \equiv 7 \pmod{40}$. As p runs over the units modulo 7, $p^2 \in \{1, 2, 4\}$, while $-20 \equiv 1$, $-40 \equiv 2$, $-80 \equiv 4 \pmod{7}$. So 7 divides one of $p^2 + 20$, $p^2 + 40$, $p^2 + 80$, and $D = 7$ is a target divisor of $(5, 1)$, $(10, 1)$ or $(5, 2)$. Finally $ck_{\min}(193) = 10$ by direct computation, and $ck_{\min}(p) \geq n_p = 5$ always. $\square$

The proof is a Type I covering of Σ_5 of height 10 with $F \in \{3, 7\}$. ES itself is classical for $p \equiv 2, 3 \pmod{5}$; the content is the bounded depth.

4.3. Compactness and Hypothesis H.

Theorem 4.2 ([14, Thm. A]). *Let $r \geq 5$ be prime.*

- (i) (PROVED) *If $X_r < \infty$, then $ck_{\min}(p) \leq X_r$ for every hard p with $n_p = r$ beyond an explicit bound, so $C^*(r) \leq X_r$.*
- (ii) (CONDITIONAL ON H) *If no Type I covering of height $\leq X$ exists, then, under Hypothesis H for an explicit finite family $\mathcal{P}_X$ of polynomials, there are infinitely many hard p with $n_p = r$ and $ck_{\min}(p) > X$.*
- (iii) (PROVED) *$X_r = \infty$ if and only if Σ_r contains a point lying in no class $\text{Cl}(\kappa)$ of a certificate (a Type I-sterile point). (CONDITIONAL ON H) Under H, $C^*(r) = X_r$; in particular $C^*(r) = \infty$ if and only if a Type I-sterile point exists.*

Proof sketch. (i) Let $\mathcal{K}$ be a covering of height X_r and $F_{\max}$ its largest F . For a prime $p > \max(2X_r, F_{\max}, r)$ that is hard with $n_p = r$, p is prime to all moduli of $\mathcal{K}$, so its residue is that of a point of Σ_r and lies in some $\text{Cl}(\kappa)$. For $ck \leq X_r < p/2$ one checks $(c, k) \in \mathcal{B}_p$, and $M_{c,k}(p) \geq 1$.

(iii) The first statement is Remark 2.7 with the height ck . The second follows from (i) and (ii).

(ii) By compactness some $x^* \in \Sigma_r$ lies in no class of height $\leq X$. Fix a prime $B \geq \max(X, r, 2m + 2)$, where m is the number of slices of height $\leq X$. At each prime $q \leq B$ at most one value $V = ck^2$ has $x_q^{*2} + 4V = 0$; perturbing x_q^* by a high power of q makes every $v_q(N_{c,k})$ finite and constant on a small class $\mathcal{C} \ni (\text{perturbed point}) \subset \Sigma_r$. Let $f_{c,k}$ be the resulting fixed B -smooth part of $N_{c,k}$. A divisor $d \mid f_{c,k}$ in the target class would give a certificate containing x^* , so none exists. The family $\mathcal{P}_X$ consists of $Qt + a$ (the class $\mathcal{C}$) and the quadratics $((Qt + a)^2 + 4ck^2)/f_{c,k}$; it has no fixed prime divisor. Under H all its members are prime for infinitely many t , and then every divisor of $N_{c,k} = f_{c,k}R_{c,k}$ is d or $dR_{c,k}$ with $d \mid f_{c,k}$; both are excluded

from the target class. So $M_{c,k}(p) = 0$ for all slices of height $\leq X$. Full details are in [14, §1]. $\square$

Forcedness is not needed in (ii): forced slices simply contain no certificate class point of Σ_r . A Type I-sterile point need not be sterile in the sense of Definition 2.3; it may lie in a class of another family. Theorem 4.2 concerns the depth $ck_{\min}$, not ES.

Theorem 4.3 (CONDITIONAL on H; [13, Thm. 2.1]). *Let $g \geq 1$, $D(g) = \sum_{m \leq g} \tau(m)$, and let $r \geq 100D(g)^2 g^3$ be prime. Under Hypothesis H for an explicit family of $1 + D(g)$ polynomials, there are infinitely many hard p with $n_p = r$ and $ck_{\min}(p) > g n_p$. Consequently, under H for a finite family depending on ε , $ck_{\min}(p) \geq n_p^{2-\varepsilon}$ for infinitely many hard p .*

The construction is the one of Theorem 4.2(ii) at the point $x_q = 1$ ($q \neq r$), $x_r = b$ with a suitable non-residue b . So no bound $ck_{\min}(p) \leq f(n_p)$ with $f(r) \leq r^{2-\varepsilon}$ for all large r can be proved without refuting H for an explicit family. For fixed r , formal evaluations at H-generic points give explicit lower bounds [13, §6.2]: under H there are infinitely many hard p with $n_p = 7$ and $ck_{\min}(p) = 539$, and with $n_p = 11$ and $ck_{\min}(p) > 3000$. Unconditionally (the same refinement argument plus Dirichlet), every Type I covering of Σ_{11} has height > 3000 (CERTIFIED by finite computation; two independent engines). For $r = 7$ these bounds are superseded by §4.4.

4.4. The sign point.

Lemma 4.4 (PROVED; [14, Lemma 2.1 and §4]). *Let $x \in \Sigma_r$ with $x_q \in (\mathbb{Z}_q^\times)^2$ for every prime $q \neq r$. Then a certificate (c, k, F) with $x \in \text{Cl}(c, k, F)$ has $v_r(c)$ odd.*

Proof. $\chi_s(x)$ is determined by $x \bmod 4s$, and equals -1 exactly when $r \mid s$. If $\chi_s(x) = 1$ and $x \in \text{Cl}(c, k, F)$, the open set $\text{Cl}(c, k, F) \cap \{\chi_s = 1\} \cap \Sigma_r$ contains primes p with $(c, k) \in \mathcal{B}_p$ and $M_{c,k}(p) \geq 1$ (Dirichlet), contradicting forcing. $\square$

Definition 4.5. For $w \in \mathbb{Z}_2$ with $w \equiv 9 \pmod{16}$ and a prime $r \equiv 3 \pmod{4}$, the *sign point* $\hat{x}_w = \hat{x}_w^{(r)} \in \widehat{\mathbb{Z}}^\times$ has components w at 2, -1 at r and 1 at every other prime. For $r = 7$ we write $\hat{x}_9$ for $w = 9$.

The sign point lies in Σ_r because -1 is a non-square modulo r ; for $r \equiv 1 \pmod{4}$ no such point exists. At $\hat{x}_w$ the odd components satisfy $\hat{x}_q^2 = 1$, so $N_{c,k}(\hat{x}) = 1 + 4ck^2$ at every odd prime, and by Lemma 4.4 a certificate at $\hat{x}_w$ is an integer solution of

$$\begin{aligned} F \mid 1 + 4ck^2, \quad F &\equiv -1 \pmod{m'}, \quad F \equiv 1 \pmod{r^{v_r(ck)}}, \\ F &\equiv -w \pmod{2^{v_2(4ck)}}, \quad v_r(c) \text{ odd}, \end{aligned} \quad (2)$$

where m' is the part of ck prime to $2r$. Write $c = 2^\alpha r^\alpha c'$, $k = 2^\gamma r^b k'$ with $c'k'$ prime to $2r$, $t = v_2(4ck) = 2 + \alpha + \gamma$, and $e = (1 + 4ck^2)/F$.

Proposition 4.6 (PROVED; [14, Lemma 3.1, Prop. 4.1]). (a) If $r \equiv 7 \pmod{8}$, no certificate at $\hat{x}_w^{(r)}$ has $F^2 = 1 + 4ck^2$.
 (b) If $r \equiv 3 \pmod{8}$, then $(r(r+1)/4, 2, 2r+1)$ is a certificate at every $\hat{x}_w^{(r)}$, of height $r(r+1)/2$.

Proof. (b) $v_r(c) = 1$, $1 + 4ck^2 = 1 + 4r(r+1) = (2r+1)^2$, the 2-part of $4ck$ is 8, and $F = 2r+1$ satisfies $F \equiv 1 \pmod{r}$, $F \equiv -1 \pmod{(r+1)/4}$, $-F \equiv 1 \pmod{8}$. (a) If $F^2 = 1 + 4ck^2$, then from (2) $m' \mid F+1$ and $r^{a+b} \mid F-1$; as $\gcd(F-1, F+1) = 2$ this gives $F+1 = 2^i c' k'^2$, $F-1 = 2^j r^{a+2b}$ with $i+j = 2 + \alpha + 2\gamma$, $\min(i, j) = 1$ and $s = a + 2b$ odd. If $j = 1$, then $2^{i-1} c' k'^2 = 1 + r^s$ and $v_2(1 + r^s) = v_2(1 + r) \geq 3$ for odd s when $r \equiv 7 \pmod{8}$, so $\alpha + 2\gamma = i - 1 \geq 3$, $\alpha + \gamma \geq 2$ and the 2-adic modulus $2^{2+\alpha+\gamma}$ is at least 16; as $r^s \equiv r \pmod{16}$ and $2r \equiv 14 \pmod{16}$, $-F = -1 - 2r^s \equiv 1 \pmod{16}$, against $w \equiv 9 \pmod{16}$; the cases $i = 1$ lead to $-F \not\equiv 1 \pmod{8}$ or to a parity contradiction. See [14, Lemma 3.1]. $\square$

So sign points survive the square family exactly for $r \equiv 7 \pmod{8}$: $r = 7, 23, 31, 47, \dots$. At the other natural candidate, the *residue-one point* ($x_q = 1$ for $q \neq 7$), every non-square x_7 is already covered by $(7, 3, 11)$, $(7, 3, 23)$ or $(14, 2, 15)$ [14, Prop. 2.3].

Lemma 4.7 (PROVED; [15, Lemmas 1.1–1.2]). Let (c, k, F) be a certificate at $\hat{x}_w$ and $f \in \{F, e\}$. Then $m' \mid f+1$, $r^{a+b} \mid f-1$, $f \equiv -w \pmod{2^t}$ if $f = F$ and $wf \equiv -1 \pmod{2^t}$ if $f = e$, and $4ck^2 \equiv -1 \pmod{f}$. Conversely such data with a odd and $a, c', k' \geq 1$ give a certificate. If w is a positive integer, each f belongs to finitely many certificates, all explicitly computable from the factorisation of $(f+1)_{\text{odd}} r^{v_r(f-1)}$. If $ck \leq X$ and $r \mid c$, then $\min(F, e) < 2X/\sqrt{r} + 1$.

The finiteness needs w with finite 2-adic depth $v_2(f+w)$, $v_2(wf+1)$: at $w = -71$ the triples $(14 \cdot 2^{35j}, 8192, 71)$ are certificates for all $j \geq 0$. For $w = 9$ the lemma reduces the search to a sieve over f , at all heights simultaneously.

Computation 4.8 (CERTIFIED; [14, Comp. 3.2], [15, Comp. 2.1, 2.3]). Exhaustive searches at the sign point give the following.

- (a) No certificate at $\hat{x}_9$ ($r = 7$) has $\min(F, e) < 10^{12}$. Two independent f -graded engines cover $f < 10^{11}$; the range $[10^{11}, 10^{12})$ rests on one engine. An earlier height-graded search ($7.6 \cdot 10^9$ slices with $ck \leq 3 \cdot 10^9$) was independently re-run to its full range, and cross-checked by a factoring engine and a complete parametric enumeration on smaller ranges. (Programs: `typei3_fsearch.c`, `review_typei3_fs.c`, `typei2_signcheck.c`.)
- (b) For $r = 23, 31, 47$ and $w = 9$ no certificate has $\min(F, e) < 10^{11}$ (two engines for $r = 23$, one for $r = 31, 47$).

On ten test points (six of them carrying certificates, 64 certificates with $ck \leq 2 \cdot 10^5$ in all) the f -graded and the height-graded engines produce identical certificate sets; at the point $w = -15$ both sets are first restricted

to certificates without the divisor 15, which has infinite 2-adic depth there [15, §2].

Corollary 4.9. (CERTIFIED) *Every Type I covering of Σ_7 has height $> 1.32 \cdot 10^{12}$; for $r = 23, 31, 47$ every Type I covering of Σ_r has height $> 2.39 \cdot 10^{11}$, $> 2.78 \cdot 10^{11}$, $> 3.42 \cdot 10^{11}$ respectively. (CONDITIONAL ON H) There are infinitely many hard p with $n_p = 7$ and $ck_{\min}(p) > 1.32 \cdot 10^{12}$; so $C(7) > 1.32 \cdot 10^{12}$, and similarly for $r = 23, 31, 47$.*

Proof. By Lemma 4.7, a certificate of height $\leq X$ at $\hat{x}_9$ has $\min(F, e) < 2X/\sqrt{7} + 1$; with Computation 4.8 there is none with $X \leq 1.32 \cdot 10^{12}$. A covering of Σ_7 must contain $\hat{x}_9$. The conditional part is Theorem 4.2(ii) with the point $x^* = \hat{x}_9$. $\square$

The previous bounds (under H) were $C(7) \geq 539$ [13] and $C(7) > 3 \cdot 10^9$ [14]; the census maximum of $ck_{\min}$ over hard $p < 10^7$ with $n_p = 7$ is 76 (EVIDENCE).

4.5. Low 2-adic levels. The certificates at the sign point can also be attacked by descent.

Lemma 4.10 (Vieta descent; PROVED; [15, Lemma 5.1]). *Let $B, K, \delta, F, e \geq 1$ be integers with $Fe = 1 + BK^2$ and $e - F = BK\delta$. Then $F \equiv e \equiv 1 \pmod{B\delta}$.*

Proof. Induction on K . If $F = 1$ then $K = \delta$ and $e = 1 + B\delta^2$. Otherwise put $\rho = K - \delta F$. From $F^2 + BK\delta F = 1 + BK^2$ we get $F^2 - 1 = BK\rho$, so $1 \leq \rho < K$. Put $F' = F - B\rho\delta$; then $F'F = 1 + B\rho^2 > 0$ and $F - F' = B\rho\delta$, and induction applies to (F', F) . $\square$

Proposition 4.11 (PROVED; [15, Cor. 5.2, Props. 5.3–5.5]). *Let $w \equiv 9 \pmod{16}$ and $r = 7$. Every certificate at $\hat{x}_w$ has $t \geq 5$ and $\alpha + 2\gamma \geq 7$.*

Sketch. By Lemma 4.7 and $w \equiv w^{-1} \pmod{16}$, $e \equiv F$ modulo $2^{\min(t, 4)}$ times the odd part of ck , and $F, e \not\equiv 1 \pmod{4}$; $F \neq e$ by Proposition 4.6(a). If $t \leq 4$, Lemma 4.10 with $B = 4c$ gives $F \equiv 1 \pmod{4}$; if $\alpha + 2\gamma \leq 4$ the same holds after the integral rescaling $c \mapsto c/4^{t-4}$, $k \mapsto 2^{t-4}k$. At levels $\alpha + 2\gamma \in \{5, 6\}$ the descent stays integral and forces $c = 2^\alpha 7^a$. At level 5 the chain of the descent, read modulo 16, only reaches divisors $\equiv \pm 1 \pmod{16}$, while certificates need $\equiv 7$; this deduction is due to the internal reviewer. At level 6 the chain is a Pell orbit; a 2-adic index condition forces its index to be an odd multiple of 3, whence $D + 3$ divides the relevant Lucas number ($D = 7^a \delta^2$), and a prime $q \mid (D + 3)/2$ gives $F \equiv 1$ and $F \equiv -1 \pmod{q}$ simultaneously. Details and machine checks are in [15, §5]. $\square$

At level $\alpha + 2\gamma = 7$ the descent leaves $\mathbb{Z}$, and 34 near misses (certificates except for the 2-adic condition) with $-F \equiv 1 \pmod{8}$ and $ck \leq 10^8$ exist there. All of them have $F \equiv 15 \pmod{16}$ [16, Obs. 1.3]; by Computation 4.13 below no certificate of level 7 has $v_7(k) \leq 7$, at any height. The level is still open.

4.6. Fibre certificates: a Pell form and an any-height bound.

The results of this subsection are from [16]; they were written after the internal referee report on this note and were reviewed separately (`reviews/pointwise-typei4-review.md`). Let $r = 7$, keep the notation $c = 2^\alpha 7^a c'$, $k = 2^\gamma 7^b k'$, $t = 2 + \alpha + \gamma$ of (2), and call $L := \alpha + 2\gamma$ the *level*. Put $c_o = 7^a c'$, $k_o = 7^b k'$ and $n = c_o k_o$. A triple (c, k, F) with a odd and $L \geq 7$ that satisfies (2) except possibly the 2-adic condition is a *fibre certificate* if $F \equiv 7 \pmod{16}$. Since $t \geq 6$, this holds if and only if (c, k, F) is a certificate at $\hat{x}_w$ for some $w \equiv 9 \pmod{16}$ (namely for every $w \equiv -F \pmod{2^t}$). Then $e \equiv F \pmod{n}$ and $v_2(e - F) = 4$, so $\delta := (e - F)/16n$ is an odd integer [16, Lemma 1.1].

Proposition 4.12 (Pell form; PROVED; [16, Prop. 1.2, Cor. 1.4]). *Let (c, k, F) be a fibre certificate of level L , oriented so that $\delta > 0$. Put $M := c_o \delta^2 + 2^{L-4}$, $d := c_o M$ and $A := (F + e)/2$. Then $A^2 - d(8k_o)^2 = 1$ and $A \equiv -1 \pmod{32}$, and there is a factorisation $M = P_1 Q_1$ with*

$$16 c' P_1 k'^2 - 7^a Q_1 \cdot 49^b = 1. \quad (3)$$

Equivalently, with $D := 7^{a+b} \delta$, $g := 4P_1 k' - D$ and $h := 4P_1 k' + D$,

$$c' gh - P_1 = 2^{L-4} 7^{a+2b}, \quad F = 8c' k' g - 1, \quad e = 8c' k' h - 1.$$

Conversely, for $L \geq 7$ (R98 repair), every solution of (3) in positive integers with $P_1 Q_1 = c_o \delta^2 + 2^{L-4}$, δ, a, c', k' odd and $7 \nmid c' k'$ gives a fibre certificate, with $F, e = A \mp 8n\delta$ and $A = 2 \cdot 7^a Q_1 \cdot 49^b + 1$, for each split $\alpha + 2\gamma = L$.

Sketch. $A^2 - ((e - F)/2)^2 = Fe = 1 + 2^{L+2} c_o k_o^2$ and $(e - F)/2 = 8n\delta$ give the norm equation. From $F \equiv 7$, $e \equiv 23 \pmod{32}$ (or vice versa) we get $A \equiv 15 \pmod{16}$, so $v_2(A - 1) = 1$ and $v_2(A + 1) = 5$. The congruences $F \equiv -1 \pmod{c' k'}$, $F \equiv 1 \pmod{7^{a+b}}$ and $F \equiv A \pmod{n}$ put $c' k'^2$ into $A + 1$ and 7^{a+2b} into $A - 1$; the factor M , prime to $7c_o$, is split between the two. Then $(A + 1)/2 - (A - 1)/2 = 1$ is (3). For the converse, $(4k' \sqrt{c' P_1} + 7^b \sqrt{7^a Q_1})^2 = A + 8k_o \sqrt{d}$ has norm 1, and the congruences of (2) and $F \equiv 7 \pmod{16}$ follow because k' and $n\delta$ are odd. The second form is a rewriting. Details are in [16, §1]; the identities are re-checked on every certificate found below by block (eb) of `verify.py`. $\square$

So a fibre certificate is a norm-one unit of $\mathbb{Z}[\sqrt{d}]$ which is the square of an element of the special shape above. The sign conditions -1 at $c' k'$ and $+1$ at 7 are built into the factorisation $(A + 1)(A - 1)$; in these coordinates quadratic residue symbols give no obstruction (ASSESSMENT, [16, §4]). The second form makes each pair (L, b) a finite problem. With $T = 2^{L-4}$ and $y := c' g \delta$, every solution has y odd, $y < T \cdot 7^b$, $P_1 = c' g^2 + 7^{a+b}(2y - T7^b)$, $P_1 \mid T7^b - y$, and $7^a < T/2$ if $2y > T7^b$, $7^a < T^2 7^b/4$ if $2y < T7^b$ [16, Lemma 3.1, Cor. 3.2] (PROVED). For each (L, b) this is a complete search over all a and all heights c, k .

Computation 4.13 (CERTIFIED; [16, Comp. 3.4, Cor. 3.5]). The search of [16, Cor. 3.2] finds no fibre certificate with $L \leq 10$ and $b \leq 7$. For $11 \leq L \leq 22$ and $b \leq 3$ fibre certificates exist exactly at $(L, b) = (11, 0)$, $(13, 1)$, $(14, 0)$ (two), $(14, 3)$, $(16, 0)$ (three), $(18, 0)$, $(18, 1)$, $(19, 0)$ (two), $(20, 0)$ (four), $(21, 0)$, $(22, 1)$ (two); at $L = 23, \dots, 26$ with $b \leq 1$, and at $L = 23, 24$ with $b = 2$, there are 17 more. In all of them $\max(v_2(F+9), v_2(e+9)) < 2 + \lceil L/2 \rceil \leq t$ (the largest value is 10, at $L = 26$). Hence no certificate at $\hat{x}_9$ has

$$(L \leq 10, v_7(k) \leq 7), \quad (L \leq 22, v_7(k) \leq 3), \\ (L \leq 24, v_7(k) \leq 2) \quad \text{or} \quad (L \leq 26, v_7(k) \leq 1),$$

at any height and for any $v_7(c)$. The author's engine (`typei4_lb.c`) covers all these ranges. A second complete engine, written independently by the reviewer in other coordinates and with its own bounds (`review_typei4_jsearch.c`), reproduces the lists exactly for $L \leq 8, b \leq 6$; $L \leq 10, b \leq 5$; $L \leq 17, b \leq 3$; $L \leq 18, b \leq 2$; $L \leq 23, b \leq 1$; and $L \leq 26, b = 0$. A third complete engine, written from scratch in a later review in the coordinates of [16, Lemma 3.1] and checked against a height-bounded brute force from the definition (`review_r98_fibre.c`), reproduces the lists exactly on all ranges of the computation, and the engine `review_typei5_relax.c` of the review of [17] finds no fibre certificate with $L \leq 10, b \leq 9$ (R98 repair).

This bound is of a different type from Computation 4.8: the height is unbounded, and the 2-adic and 7-adic depths are bounded. The two results complement each other.

Remark 4.14 (Scope; CERTIFIED). For $L \leq 10$ the fibre is empty in the range searched. For each $L \in \{11, 13, 14, 16, 18, 19, 20, 21, 22\}$, Computation 4.13 gives a certificate of level L at some $\hat{x}_{w'}$ with $w' \equiv 9 \pmod{16}$. For example, $(c, k, F) = (42, 32, 71)$ has $1 + 4ck^2 = 172033 = 71 \cdot 2423$, $t = 8$, $c' = 3$ and $L = 11$. It is a certificate at $\hat{x}_w$ for every $w \equiv 185 \pmod{256}$, but not at $\hat{x}_9$, since $v_2(71 + 9) = 4$ and $v_2(2423 + 9) = 7$ are both < 8 . So at these levels it is false that no $\hat{x}_w$ with $w \equiv 9 \pmod{16}$ carries a certificate of level L [16, Prop. 4.1]. Consequently (ASSESSMENT), an argument that uses w only modulo 16, such as the proof of Proposition 4.11, cannot reach these levels; there the exact value $w = 9$ must enter at depth $\geq t$.

The open part is therefore the 7-adic tower $b \rightarrow \infty$ at the levels $L = 7, \dots, 10$. At $L = 7$ put $u = 7^b$, $j := 4u - y$ (odd, $1 \leq j < 4u$) and $m := c'\delta^2$. Then for $7 \nmid j$ the integer $\lambda := (\rho j - m)/u$, where $\rho := (4u + j)/P_1$, is even with $2 \leq \lambda \leq 4j + j^2/u$, and $7^{b-a} < 2\lambda j^3 + 10j^2 + 37j$. So for each fixed j prime to 7 only finitely many b occur, and $j = 1$ occurs for no b [16, Lemma 3.6] (PROVED). The integrality of λ holds without the hypothesis $7 \nmid j$, and at every level $L \geq 5$, if one puts $j := T7^b/2 - y$: modulo u one has $\rho P_1 \equiv j$, $m P_1 \equiv j^2$ and $7 \nmid P_1$ [17, Lemma 3.1]. The following results of [17] (reviewed in `reviews/pointwise-typei5-review.md`) treat the levels $L = 7, \dots, 10$ uniformly (R98 repair).

Proposition 4.15 (PROVED; [17, Lemma 1.1, Cor. 1.2]). *Let $P, Q \geq 1$ be integers with $7 \mid Q$ and PQ not a square, and let (X_1, u_1) be the solution of $16PX^2 - Qu^2 = 1$ in positive integers with u_1 least. If some solution has u odd, then no positive solution other than (X_1, u_1) has u a power of 7. Consequently a fibre certificate is determined by (a, c', δ, P_1) , with $7^b = u_1$, and its unit $A + 8k_o\sqrt{d}$ is ε or ε^2 , where ε is the fundamental norm-one unit of $\mathbb{Z}[\sqrt{d}]$.*

Sketch. Put $\alpha = 4X_1\sqrt{P} + u_1\sqrt{Q}$. The positive solutions are the α^n with n odd: α^2 generates the positive part of the group $\{A + 4B\sqrt{d} : A^2 - 16dB^2 = 1\}$, since otherwise there would be a smaller solution, or P would be a square and every u even. Write $\alpha^n = 4X_n\sqrt{P} + u_n\sqrt{Q}$. Then $u_n = u_1L_n$ with integers $L_n \equiv n \pmod{7}$, $L_7 \equiv 7 \pmod{49}$, $L_7 > 7$, and $L_{mn}(\alpha) = L_n(\alpha)L_m(\alpha^n)$. Writing $n = 7^r n'$ with $7 \nmid n'$ shows that L_n is a power of 7 only for $n = 1$. This is a routine lifting-the-exponent argument for the Lucas sequence u_n/u_1 (O101 audit). For the certificate apply this to $P = c'P_1$, $Q = 7^a Q_1$ of Proposition 4.12; $d = PQ$ is not a square as $v_7(d) = a$ is odd. Details are in [17, §1]. $\square$

In particular b is a function of the field data, and an engine that runs over d and tests ε and ε^2 is complete for all b . Such an engine, which needs ε only modulo 2^{64} and 7^{22} along the continued fraction of $\sqrt{d}$, is `typei5_dmod.c` [17, §2].

Theorem 4.16 (PROVED and CERTIFIED; [17, Prop. 3.3, Lemma 3.6, Thm. 3.7]). *Let $L \in \{7, 8, 9, 10\}$, put $u = 7^b$, $j := Tu/2 - y$, $m := c'\delta^2$, $\rho := (Tu - y)/P_1$, $\lambda := (\rho j - m)/u$ and $\Delta := 8j \cdot 7^a m + 6Tj - T^2u$. A certificate at $\hat{x}_9$ of level L satisfies all of the following.*

- (a) $2y < T7^b$ (PROVED, by hand).
- (b) $\lambda \geq 1$ and $\Delta > 2Tj$. (PROVED that $\lambda = 0$ is impossible and that $\Delta = 2Tj$ forces $L = 9$, $b = 0$, which Computation 4.13 excludes; PROVED that the regimes $\lambda < 0$ and $\Delta < 2Tj$ are finite at each L , and CERTIFIED that they are empty by a complete computation, reproduced independently in review.)
- (c) $v_7(k) \geq 8$ (CERTIFIED, Computation 4.13 and two further complete engines); $v_7(k) \geq 9$ by two engines and $v_7(k) \geq 10$ by one.
- (d) $c_o\delta > 10^6$ (CERTIFIED by `typei5_dmod.c`, valid for all b by Proposition 4.15; one engine, replayed in review).

In the remaining regime, with $\sigma := \Delta - 2Tj \geq 1$, one has $7^b < 4j^2/\sigma$, and for each fixed (σ, a, λ) there are finitely many explicit candidates [17, Prop. 3.3(v)]. What is open is that σ and λ are unbounded. By Proposition 4.15 this asks whether 7^b can be the $\sqrt{Q}$ -coefficient of the minimal solution for a two-parameter family of fields $d = c_o(c_o\delta^2 + T)$. Linear forms in logarithms bound exponents in a fixed field or a fixed recurrence, so they do not apply directly, and reciprocity gives no obstruction either (ASSESSMENT, [17, §3]).

4.7. Scope of a sterility proof.

Proposition 4.17 (PROVED; [15, Prop. 3.1]). *The set of Type I-sterile points of Σ_7 is closed and has empty interior.*

Proof. Closedness is Remark 2.7. Let $U = \{y \equiv x \pmod{Q}\} \subset \Sigma_7$ be a basic neighbourhood, $Q = 2^j 7^i \prod_{q \in P} q^{e_q}$. Pick a prime $p_0 \notin P \cup \{2, 7\}$ and, by Dirichlet, a prime $F \notin P \cup \{2, 7, p_0\}$ with $F \equiv 3 \pmod{4}$, $F \equiv -x_7 \pmod{7}$ and $(-7p_0/F) = 1$; these are compatible because $(-7/F) = (F/7) = (-x_7/7) = 1$ for non-square x_7 . Then $\text{Cl}(7p_0, 1, F)$ meets U by the Chinese remainder theorem. $\square$

So no sterility proof can consist of exhibiting one ambient congruence cylinder: a sterile point, if it exists, is a limit of covered points. This does not exclude arguments within the thin sign fibre $\Phi = \{\hat{x}_w : w \in 9 + 16\mathbb{Z}_2\}$, which itself has empty interior.

Remark 4.18 (Measure route; PROVED reduction and EVIDENCE). Theorem 4.2(iii) needs *some* Type I-sterile point, not $\hat{x}_9$. By Lemma 4.7, the certificates having a given $f \equiv 7 \pmod{16}$ remove from Φ at most two 2-adic balls of radius $2^{-t_{\min}(f)}$, with $t_{\min}(f) \geq 5$ by Proposition 4.11; other f remove nothing. Let U_Y be the part of Φ surviving all $f < Y$. If

$$\mu(U_Y) > \sum_{f \geq Y} 2^{5-t_{\min}(f)}$$

(Haar measure on w , normalised to 1), then Φ contains a Type I-sterile point, and $C^*(7) = \infty$ under H. Numerically $\mu(U_{10^{10}}) \geq 0.6086$, the bound including the error of truncating the 2-adic depth at 40; and $t_{\min}(f)$ grows like $\frac{1}{2} \log_2 f$ in the data (EVIDENCE). An explicit bound for the number of f with given $t_{\min}$ that makes $\sum_{f \geq 10^{10}} 2^{5-t_{\min}(f)} < 0.6$ would therefore close the question; summability alone does not suffice (the tail must beat the computed $\mu(U_Y)$), let alone a bound $o(2^T)$. No such bound is known.

Conjecture 4.19 ([14, Conj. 3.4]). *The sign point $\hat{x}_9$ is Type I-sterile. Then, under H, $C(7) = C^*(7) = \infty$.*

The evidence is the search range of Computation 4.8, the empty levels of Proposition 4.11, and the decay of the near-miss mass (EVIDENCE). Among integral points $(x_2, x_7) = (w, z)$ with $w \equiv 9 \pmod{16}$, $|w| \leq 200$, $|z| \leq 60$ (all other components 1), the 26 that survive the complete enumeration of certificates at $\{2, 7\}$ -generic points up to the cutoff of [14, §2.2] (parameter $\Lambda \leq 3 \cdot 10^8$ there) all have $z = -1$; this is how the sign point was found. For $r \equiv 1 \pmod{4}$ no sign point exists, and $r = 5$ is the one value with a known finite covering.

5. MORDELL-TYPE COVERINGS MODULO A FURTHER PRIME

We now return to all seven families and to ES itself. For a prime $r \geq 11$ let $\Sigma_r \subset \hat{\mathbb{Z}}^\times$ (the *main variant*) be the set of units x with $x \equiv 1 \pmod{24}$,

x a square modulo 5 and 7 (Mordell-hard) and x a non-square modulo r , and let $\Sigma_r^{\text{np}} \subset \Sigma_r$ also require x to be a non-zero square modulo every prime $5 \leq \ell < r$. By reciprocity the primes $p > r$ in Σ_r^{np} are the Mordell-hard primes with $n_p = r$. (Throughout this section Σ_r denotes this set, not the set of §4.)

5.1. A finite-exception theorem for $r = 13$.

Theorem 5.1 (PROVED by finite computation; [18, Thm. 3.1]). *Let p be a prime with $(p/13) = -1$.*

- (a) *If $(p/11) = +1$, then $4/p = 1/x + 1/y + 1/z$ has a solution in positive integers unless $p \equiv 112561 \pmod{240240}$, i.e. $p \equiv 1 \pmod{16 \cdot 3 \cdot 5 \cdot 7}$, $p \equiv 9 \pmod{11}$, $p \equiv 7 \pmod{13}$.*
- (b) *Without a condition at 11, the same holds unless $p \bmod 720720 \in \{112561, 352801, 380881, 418321, 473761, 483841\}$.*
- (c) *If $(p/11) = +1$, the exceptions of (a) shrink to $p \bmod 720720 \in \{112561, 352801\}$.*

Proof. If p is not a square modulo 840, Mordell’s identities apply. Otherwise $p \bmod L$ represents a point of Σ_{13}^{np} ($L = 240240$) or of Σ_{13} ($L = 720720$). Explicit lists of 20 resp. 31 ET classes, with moduli dividing L , cover every such residue except the listed ones; Lemma 2.1 then gives the solution for every $n \geq 1$ in the class. Part (c) combines the two lists: the third lift 593041 of 112561 modulo 720720 is covered by the second list. The certificates are the files `cert_r13_np_240240.json` and `cert_r13_main_720720.json` in `data/mordell/`; the stand-alone checker `mordell_check.py` verifies the polynomial identities, positivity and the integer-valuedness of all coordinates on every class residue. An independent checker written from scratch in review (`review_mordell_check.py`) confirms both certificates. $\square$

Novelty. Theorem 5.1 is an explicit packaging of the level sieve of Salez [21, §§3–4], extended by Mihnea and Dumitru [7]. Salez sieves with seven modular equations, including filters with composite moduli, and tabulates the residues not certified at levels 840, 9240, 120120, $\dots$; using all ET classes of modulus dividing L reproduces his counts 34 (at 9240) and 192 (at 120120) exactly. Salez’s residue set modulo 120120 already contains only 7 classes with $(p/13) = -1$, so a statement of the shape of Theorem 5.1 modulo 120120 follows from his 2014 data. Theorem 5.1 is the same sieve one level deeper ($2^4, 3^2$), sliced by $(p/13)$, with certificates that can be checked independently. Finite-exception statements involving 11 and 13 are older still: Terzi [24] lists 198 exceptional residues modulo 120120, and Ionascu and Wilson [5] carry the analysis to the moduli 9240 and $2762760 = 2^3 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 23$ (O101 audit; [24] only via [5]). The exceptional classes carry $1/360$ (a) and $6/2160$ (b) of the relevant residues; deeper levels shrink them (to 1412 residues modulo $2^4 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13^2 \cdot 17 \cdot 19 \cdot 23$ with moduli $\leq 10^8$ in case (b)). Whether they ever reach zero is open: the point x^* below was a candidate obstruction but is covered (Proposition 5.4).

5.2. Generic points and boxes. For a finite set T of primes call $x \in \widehat{\mathbb{Z}}^\times$ T -generic if $x_q = 1$ for every $q \notin T$. A class of modulus $M = M_T N$, with M_T the T -part, meets the T -generic points exactly when its residue satisfies the class condition at $n = 1$ away from T ; its trace on the T -generic points is then a box (a finite union of products of balls) in $\prod_{q \in T} \mathbb{Z}_q$. In ET's coordinates this says that the non-constant coordinates of the family are $\mathbb{Z}[1/T]$ -integral at $n = 1$ [18, §2.1]. For the families II1, I4 and II2 the conditions take rigid forms, e.g. II1 and I4 at a T -generic point amount to $4iabk = F(a + b + k)$ with $k = (a + b)/e$, $e + 1 = 4iab/F$ and F the T -part of $4ab$; so for each T -level F there are finitely many boxes (PROVED; the II1, I4 and II2 rows were re-derived in the review of [18], and the full table for $T = \{17\}$, all seven families, in the review of [20]).

At the 13-generic points of Σ_{13} every non-residue class is covered at 13-adic level one: $x_{13} \equiv 2$ by II2 $(a, d, f) = (9, 2, 143)$, $x_{13} \equiv 7$ by II2 $(2, 2, 143)$, and $x_{13} \equiv 5, 6, 8, 11$ by II2 classes with $f = 39$ [18, §1] (the cases $x_{13} \equiv 2, 5, 7$ were re-checked in review). So hard points for $r = 13$ need special residues at 11 as well. For $T = \{11, 13\}$ the classes of modulus $\leq 10^6$ leave uncovered only cells above $(x_{11}, x_{13}) \equiv (2, 2) \pmod{(11, 13)}$ (EVIDENCE). The first layer of this is in Yamamoto's Table 1 [25, p. 44]: the non-residue classes modulo q that his simplified coverings of level q leave open are $2, 6 \pmod{11}$, $2, 7 \pmod{13}$ and $3, 5, 6, 7 \pmod{17}$. The residues $x_{11} \equiv x_{13} \equiv 2$ are among them, and so are the residues modulo 13 of all six exceptional classes of Theorem 5.1. The joint cell $(2, 2)$ and the profinite points above it are not treated there (O101 audit).

5.3. The point x^* and the candidate x^{**} .

Computation 5.2 (CERTIFIED; [18, Comp. 4.1, §4.1]). Let $x^* \in \widehat{\mathbb{Z}}^\times$ have $x_{11}^* = x_{13}^* = 2$ and $x_q^* = 1$ for every other prime q . Then $x^* \in \Sigma_{13}$ and

- (a) x^* lies in no ET class of modulus $\leq 10^6$ (two independent engines, `mordell_point.py` and the reviewer's `review_mordell_point.py`);
- (b) x^* lies in no II1, I4 or II2 class whose modulus has $\{11, 13\}$ -part dividing $11^4 \cdot 13^4$, with unrestricted remaining part (complete rigid enumeration; re-checked by a second engine for every level dividing $11^3 \cdot 13^3$ except $11^3 \cdot 13^3$ itself; that level and the levels with exponent 4 rest on one engine).

Corollary 5.3 (PROVED from Computation 5.2(a)). *Every finite set of ET classes containing all but finitely many Mordell-hard primes p with $(p/13) = -1$ contains a class of modulus $> 10^6$. The same holds for the primes with $(p/11) = (p/13) = -1$.*

Proof. Remark 2.6: x^* is an accumulation point of these primes by Lemma 2.4 $((2/11) = (2/13) = -1)$. $\square$

An earlier version of this note conjectured, following [18, Conj. 4.2], that x^* is sterile. That conjecture is false.

Proposition 5.4 (PROVED; [19, Thm. 3.1]). *The point x^* lies in the II3 class with $(a, d, e) = (8, 33, 11999)$, that is, $n \equiv 12650497 \pmod{12670944}$, where $12670944 = 4ade = 2^5 \cdot 3 \cdot 11 \cdot 13^2 \cdot 71$. It also lies in the I2 class with $(a, c, f) = (125, 88, 11999)$, of modulus 527956000. In particular x^* is not sterile.*

Proof. Write $M = 4ade = M'M_T$ with $M_T = 11 \cdot 13^2$ and $M' = 6816$. The residue $r = -4a^2d - e \pmod{M}$ equals 12650497. We have $r \equiv 1 \pmod{M'}$ and $r \equiv 2 \pmod{M_T}$; equivalently, $e + 1 = 12000 \equiv 0 \pmod{96}$, $11 \mid e + 2 = 11 \cdot 1091$, $71 \mid 4a^2d + 1 = 71 \cdot 119$, and $13^2 \mid 2a^2d + 1 = 25 \cdot 13^2$. So the clopen class contains x^* . The side condition $(4ad, e) = (1056, 11999) = 1$ holds, and Lemma 2.1 applies. For example, $p = 12650497$ is prime, and the II3 solution is $4/p = 1/3165624 + 1/3339731208 + 1/5005839614391$. $\square$

Computation 5.2 is not affected: both classes have modulus $> 10^6$, and they belong to the families II3 and I2, which are outside the rigid enumeration (b). The covering question for the primes with $(p/11) = (p/13) = -1$ is therefore open again. In [19] every class meeting the $\{11, 13\}$ -generic points is attached to a solution of $4/N = 1/x + 1/y + 1/z$ with N an $\{11, 13\}$ -unit; this makes complete enumeration by N possible. The (II3, I2) pair above has $N = 11 \cdot 13^2$. Among the $\{11, 13\}$ -generic points above the cell $(2, 2)$, the enumeration with $N \leq 4 \cdot 10^7$ leaves 15 of 143 subcells modulo $(11^2, 13^2)$ uncovered. A targeted search was run at four points of small height in these subcells. One of them is the point x^{**} with $x_{11}^{**} = 2$, $x_{13}^{**} = 15$ and $x_q^{**} = 1$ for every other prime q ; it also lies in Σ_{13} .

Computation 5.5 (CERTIFIED within the stated ranges; [19, Comp. 5.1, 5.3]). The point x^{**} lies in no ET class of family II3, I3, I1 or II2 with e (resp. f) $\leq 10^9$, and in no class of family I2, II1 or I4 with f (resp. e) $\leq 2 \cdot 10^7$ and $\{11, 13\}$ -exponents of absolute value at most 12. For II3, I3, I1 and II2 these ranges impose no restriction on the $\{11, 13\}$ -part of the modulus: the box conditions force $v_q(ad) \leq v_q(e + x_q^{**}) \leq 8$ for $q \in \{11, 13\}$ (II2: $v_q(ad) \leq v_q(f + 1) \leq 8$), within the engine's exponent cap. Engines: `m13b_target.c` (all four families, $e \leq 10^9$) and `m13b_target2.c` (I2, II1, I4). The ranges $e \leq 10^8$ for II3, I3, I1 were reproduced by an independent engine in the review of [19], and the full range $e, f \leq 10^9$ for all four families by a further engine written from scratch from the ET class definitions, without discrete logarithms or exponent cap (`review_r98_xss.c`, R98 repair). The ranges for I2, II1 and I4 rest on one engine.

Conjecture 5.6 (EVIDENCE only; [19, Conj. 5.2]). *The point x^{**} is sterile.*

By Proposition 2.5(b), Conjecture 5.6 *implies* that no finite set of primitive classes solvable by polynomials contains all but finitely many Mordell-hard primes with $(p/11) = (p/13) = -1$. So the method of Theorem 5.1(b), a finite list of polynomial classes, could not be extended to cover all such primes up to finitely many exceptions. This would say nothing about whether ES holds for these primes. The conjecture is not equivalent to the non-existence

of such a covering (Remark 2.6). The support for it is weak. The point x^* was covered at $e \approx 1.2 \cdot 10^4$, whereas the search at x^{**} goes almost five orders of magnitude further in e (R98 repair). No tail bound of the kind in Theorem 6.4 is available here.

*Why the odd-square obstruction would not explain x^{**} .* The point x^{**} is not square-mimicking, since x_{11}^{**} and x_{13}^{**} are non-residues. Hence Proposition 3.1 and Cited Theorem 3.2 do not apply (PROVED, trivially). Jacobi-parity arguments of the kind in [4, Prop. 1.6] force only odd total $\{11, 13\}$ -valuations of the relevant parameters, which can be satisfied ([19, Lemma 1.2] works this out at x^*). The case of x^* is a warning here. A point can escape every class up to a large modulus through rigidity and still be covered.

Data (EVIDENCE). With all-family boxes of modulus $\leq 10^6$ and the rigid II boxes, 24.8% of the $(2, 2)$ cell remains uncovered at resolution $11^4 \cdot 13^4$. The classes of ES level $N \leq 4 \cdot 10^7$ alone leave 4.64%. This is an upper bound, since that enumeration is complete only for $\{11, 13\}$ -levels ≤ 6324 [19, §4]. For the np variant no T -generic obstruction with $|T| \leq 3$ was found; a best-first tree search over Σ_{13}^{np} stopped at 30000 nodes with open Haar mass $4.1 \cdot 10^{-7}$, so that variant is undecided.

6. THE 17-GENERIC LINE

For $r = 17$ the candidate sterile set is one-dimensional, and the classes meeting it can be described exactly. A point $x \in \widehat{\mathbb{Z}}^\times$ is 17-*generic* if $x_q = 1$ for all $q \neq 17$; write $x = x(u)$ with $u = x_{17} \in \mathbb{Z}_{17}^\times$. If u is a non-residue modulo 17 then $x(u) \in \Sigma_{17}^{\text{np}}$, and by Lemma 2.4 the Mordell-hard primes with $n_p = 17$ accumulate at $x(u)$. The non-residue cells $C_5 = \{u \equiv 5\}$ and $C_7 = \{u \equiv 7 \pmod{17}\}$ are exchanged by $u \mapsto 1/u$ ($5 \cdot 7 \equiv 1$). Both residues are among the classes 3, 5, 6, 7 (mod 17) left open at level 17 in Yamamoto's Table 1 [25, p. 44] (O101 audit). A class of modulus $17^k N$ with $17 \nmid N$ contains $x(u)$ if and only if its residue ρ satisfies $\rho \equiv 1 \pmod{N}$ and $u \equiv \rho \pmod{17^k}$; it then contributes the *box* $\rho + 17^k \mathbb{Z}_{17}$ of *level* k . We use Haar measure μ on $\mathbb{Z}_{17}$, so $\mu(C_v) = 1/17$. All results of this section are from [20] and its review.

6.1. Box types.

Lemma 6.1 (PROVED; [20, Lemmas 1.1, 1.3]). *At level $F = 17^k$ the boxes of the seven families are exactly the boxes of the following types (all parameters positive integers, ' the 17-free part):*

- (P) (a, d, f) with $17 \nmid f$, $f \mid 4a^2d + 1$, $4(ad)' \mid f + 1$, $(ad)_{17} = F$; *box* $-f \pmod{F}$ (from I1, I3 with $17 \mid cd$, II3 with $17 \mid ad$);
- (Q) (a, d, f) with $f = Fg$, $17 \nmid g$, $4ad \mid f + 1$, $g \mid 4a^2d + 1$; *box* $-4a^2d \pmod{F}$ (II2, II3 with $17 \mid e$), and its inverse $-1/(4a^2d)$ (I2 with $17 \mid f$);
- (U) (a, b, e) with $e \mid a + b$, $(e, 4ab) = 1$, $4(ab)' \mid e + 1$, $(ab)_{17} = F$; *box* $-e \pmod{F}$ (III, I2 with $17 \mid ac$), and its inverse $-1/e$ (I4).

The type I3 with $17 \mid f$ (boxes $\pm\sqrt{-4a^2d}$) never meets the 17-generic line. (Q)-data exist only at odd levels k , with $-4a^2d$ a non-residue modulo 17; the set of (P)-boxes is closed under inversion; no class contains the whole line (no level-0 box).

Proof of the parity statements. For a (Q)-datum write $f = 4adm - 1 = 17^k g$. As $f \equiv -1 \pmod{4d}$, Jacobi reciprocity gives $(d/f) = 1$, and $(-1/f) = -1$, so $(-d/f) = -1$. As $g \mid 4a^2d + 1$, $-d(2a)^2 \equiv 1 \pmod{g}$, so $(-d/g) = 1$. Hence $(-d/17)^k = -1$: k is odd and $-d$, hence $-4a^2d$, is a non-residue modulo 17, so $n^2 \equiv -4a^2d \pmod{17^k}$ has no root. A level-0 box would give positive integers with $4/1 = 1/x + 1/y + 1/z$, by the rigid forms [20, Cor. 2.4], which is impossible. The type classification is read off from the generic-point conditions family by family; see [20, §1]. $\square$

6.2. Boxes are Erdős–Straus solutions at prime powers. In ET's coordinates, Σ_n^I is given by $4abcd = n(a+b) + c$ and Σ_n^{II} by $4abcd = a + b + nc$, with $e = (a+b)/c$ and the further relations of [4, §2].

Proposition 6.2 (PROVED; [20, Lemmas 2.1–2.3, 5.1, 5.2]). (a) Put $b = (f+1)/(4ad)$. Then $(a, d, f = 17^k g)$ is a (Q)-datum if and only if $(a, b, (a+b)/g, d)$ is an $\mathbb{N}$ -point of $\Sigma_{17^k}^I$ with $e = g$ prime to 17. The box is $-4a^2d \equiv -a/b \pmod{17^k}$.
 (b) A (U)-datum (a, b, e) of level k gives the solution $4/17^k = 1/(iab) + 1/(iac) + 1/(ibc)$, $c = (a+b)/e$, $i = (e+1)F/(4ab)$. Every (U)-box meeting $C_5 \cup C_7$ lies in a (P)-box of level $\leq k$ with the same centre, so the type (U) adds no measure.
 (c) Let (a, d, f) be a (P)-datum of level k , $a = 17^\alpha a'$, $d = 17^\delta d'$, $n = a'd'$, $f = 4ni - 1$, $(4a^2d + 1)/f = 4nj - 1$. Then (i, j, a', d') is an $\mathbb{N}$ -point of $\Sigma_{17^K}^{II}$ with $K = k + \alpha \in [k, 2k]$ and $17 \nmid a'd'$. Conversely every $\mathbb{N}$ -point of $\Sigma_{17^K}^{II}$ with $17 \nmid cd$ yields (P)-boxes, nested, the largest of level $\lceil K/2 \rceil$, with centre $-f_{\text{ET}}$; if $17 \nmid b$ (as for every box meeting the non-residue cells) the centre is $\equiv -a/b$. $\Sigma_{17^K}^{II}$ has no such $\mathbb{N}$ -points for even K .

Proof of (c). Since $(ad)_{17} = F = 17^k$ and $(ad)' = n$, we have $4a^2d = 4aFn$, so $(4ni - 1)(4nj - 1) = 4aFn + 1$; expanding gives $4nij - i - j = aF = 17^K a'$, i.e. $4ij a' d' = i + j + 17^K a'$, which is the defining equation of $\Sigma_{17^K}^{II}$ for (i, j, a', d') . The converse uses $4c^2dN + 1 = f(4bcd - 1)$ [4, (2.21)]. The centre is $-a/b$ because $bf = Nc + a$ [4, (2.20)]. Even K are excluded by the reciprocity argument of Lemma 6.1 applied to $f \mid 4 \cdot 17^K a'^2 d' + 1$, $f \equiv -1 \pmod{4d'}$. $\square$

So the boxes on the 17-generic line are balls around the rationals $-a/b$ built from solutions of $4/17^K = 1/x + 1/y + 1/z$: Type I points at full level for (Q), Type II points at half level for (P). The half level is what makes (P) critical below.

6.3. Exact low levels.

Computation 6.3 (CERTIFIED; [20, Comp. 3.1]). The boxes of level ≤ 5 (and the (Q), (U) boxes of level 7) were enumerated completely, using the bounds of [4, Lemma 2.8], by two independent engines (`m17_enum.c` and the reviewer's `review_m17_enum.c`); the union in C_5 was computed exactly (two 17-adic balls are nested or disjoint). The covered fraction of C_5 is

level k	boxes meeting C_5	covered fraction after level k
1	0	0
2	4 (P)	0.235294
3	32	0.314879
4	50 (P)	0.321799
5	158	0.322793
7 (Q, U only)	240	0.322797

and identically for C_7 . So the boxes of level ≤ 5 leave exactly the fraction $56561/83521 = 0.67720\dots$ of each cell uncovered. Completeness was also checked against a brute force that reads the classes literally off [4, Prop. 1.9] for all moduli $\leq 10^6$ and all boxes of level ≤ 3 (168 boxes in all cells, 0 missing); the boxes of levels 4 and 5 rest on the two enumeration engines only. Moreover the point $x(5)$ lies in no ET class of modulus $\leq 10^6$ (two engines) and in no II1, I4 or II2 class with 17-part dividing 17^4 [18, Comp. 5.1].

The numbers of data are $D_P(K) = 2, 32, 121, 258, 604$ for $K = 1, 3, 5, 7, 9$ and $D_Q(k) = 2, 73, 245, 707$, $D_U(k) = 4, 68, 310, 826$ for $k = 1, 3, 5, 7$; the measure added per level is 0.235, 0.080, 0.0069, 0.0010 at levels 2–5 (EVIDENCE of decay, from three ratios only).

6.4. The tail. Let $B_k = 4D_Q(k) + 2D_U(k) + 2D_P(2k - 1)$ be the resulting bound for the number of boxes of level k (two orientations; $D_P(2k) = 0$).

Theorem 6.4 (PROVED; [20, Thm. 4.1]). *If*

$$\sum_{k \geq 6} 17^{1-k} B_k < \frac{56561}{83521},$$

then C_5 and C_7 contain sterile points u : $x(u)$ lies in no ET class.

Proof. The boxes of level ≥ 6 cover at most $\sum_{k \geq 6} B_k 17^{-k}$, less than the measure $\frac{56561}{83521} \cdot \frac{1}{17}$ left by the levels ≤ 5 . $\square$

Corollary 6.5 (CONDITIONAL on the hypothesis of Theorem 6.4). *No finite set of primitive residue classes solvable by polynomials contains all but finitely many of the Mordell-hard primes with $n_p = 17$ (primes $p \equiv 1 \pmod{24}$) with $(p/17) = -1$ and $(p/q) = +1$ for $q = 5, 7, 11, 13$.*

Proof. Proposition 2.5(b) with the sterile point $x(u)$. $\square$

Conjecture 6.6 ([20, Conj. 4.3]). *The cell C_5 (equivalently C_7) contains a sterile point.*

Status of the hypothesis. The series converges (PROVED, non-explicitly): the proof of [4, Prop. 1.7] counts $\mathbb{N}$ -points and gives $D_Q(k) \ll 17^{(3/5+o(1))k}$ and $D_P(K) \ll 17^{(2/5+o(1))K}$, and the bound $f(n) \ll_\varepsilon n^{2/3+\varepsilon}$ of Browning and Elsholtz [2] handles (U). The constants are not explicit, so this gives no level the computation reaches. The critical family is (P): its boxes have weight $\asymp N^{-1/2}$ at ES level $N = 17^K$, against the count $N^{2/5+o(1)}$, a margin of only 1/10; even $D_P(K) \leq 17^{2K/5}$ with constant 1 would leave a tail ≈ 0.84 from $K \geq 11$, more than 0.677. An $\mathbb{N}$ -point of Σ_N^{II} with $a \leq b$ is determined by (a, b) , so it would suffice (ASSESSMENT: this is the natural route, not a proved necessity) to have an explicit bound

$$\#\{(a, b) : a \leq b, ab \leq 17^K, (-17^K \bmod 4ab) \mid a + b\} \leq C \cdot 17^{(1/2-\delta)K} \quad (4)$$

with explicit C and $\delta > 0$, together with weaker explicit bounds for (Q), (U). The data fit $D_P(K) \approx K^3$ (2, 32, 121, 258, 604 against 1, 27, 125, 343, 729; EVIDENCE), far below what is needed: $D_Q + D_U \leq 17^{k/2}$ for $k \geq 8$ and $D_P(K) \leq 17^{K/4}$ for $K \geq 11$ would make the tail < 0.01 .

Assessment 6.7 (Discrete logarithms). With $(s, t) = (c, d)$ the (P)-data of level K are exactly the (a, s, t) with $17 \nmid st$, $f = 4ast - 1 \mid s \cdot 17^K + a$ and $b = (s17^K + a)/f \geq a$ [20, §6]. As $(s, f) = 1$ the condition is $17^K \equiv -a/s \pmod{f}$: a triple with $(a, f) = 1$ is ever admissible if and only if $-a/s \in \langle 17 \rangle \subset (\mathbb{Z}/f)^\times$, and then for the K in one class modulo $\text{ord}_f 17$ that satisfy the size condition $4a^2t \leq 17^K + 2a/s$. If 17^K is modelled as equidistributed modulo f , $E[D_P(K)] \approx \sum 1/(4ast) \asymp K^3$, as observed. So (4) is an Artin-type equidistribution question for one base and many moduli. Divisor-bound methods (ET, Lenstra’s bound on divisors in residue classes [6], Coppersmith–Howgrave–Graham–Nagaraj [3], explicit divisor bounds of Nicolas–Robin type [9]) treat 17^K as an arbitrary integer; they give $N^{2/5+o(1)}$ non-explicitly and $O(N)$ explicitly, and we see no way for them to reach $N^{1/2-\delta}$ explicitly. A digit-set (Cantor) construction that avoids boxes with small integer centres was tested and fails: the normalised centres of new boxes have maxima 0.86–0.99 at levels 3–7.

7. OPEN PROBLEMS

Problem 1. Is there a sterile point in $\widehat{\mathbb{Z}}^\times$ that is not a square point? Equivalently, is there a clopen set $\Sigma \subset \widehat{\mathbb{Z}}^\times$ without square points such that no finite set of polynomial identities covers all but finitely many of the primes $\mathcal{P}(\Sigma)$? (The square points form a closed set, so a non-square sterile point has a clopen neighbourhood Σ without square points; conversely a clopen Σ is defined modulo some M . Apply Lemma 2.4 and Proposition 2.5.) Each of Conjectures 5.6 and 6.6 would give one; Conjecture 4.19 would give a Type I analogue. The integer-level expectations of [25, p. 47] and [5] point the other way but do not decide the question (§1; O101 audit).

Problem 2. Prove Conjecture 4.19, or find a certificate at $\hat{x}_9$. Two routes are visible. The first is the 7-adic tower of §4.6. By Proposition 4.12 each pair (level L , $v_7(k) = b$) is a finite computation at all heights, so what is open is $b \rightarrow \infty$ at the levels $L = 7, \dots, 10$; by Theorem 4.16 only the regime $\Delta > 2Tj$ with unbounded (σ, λ) remains there (R98 repair). At the levels $L \geq 11$, where the fibre is inhabited (Remark 4.14), the exact value $w = 9$ must be used. The second route is an explicit bound for the near-miss count of Remark 4.18, small enough that the tail beats $\mu(U_{10^{10}})$, which would give a Type I-sterile point of the sign fibre (not necessarily $\hat{x}_9$).

Problem 3. Is $X_r < \infty$ for any prime $r \geq 7$? The project’s weakly supported guess is that $r = 5$ is the only value of n_p with a finite Type I covering. For $r \equiv 1 \pmod{4}$ there is no sign point; $r = 13$ is the first open case of that kind.

Problem 4. Prove the explicit count (4) for some $\delta > 0$ and explicit C , or any other bound that makes the series of Theorem 6.4 smaller than $56561/83521$. By the Assessment of §6 this is a question about how often $-a/s$ lies in the subgroup generated by 17 modulo $4ast - 1$, and where its discrete logarithm falls.

Problem 5. Decide whether x^{**} is sterile (Conjecture 5.6), and whether the np variant Σ_{13}^{np} has a sterile point at all. A finite-level reduction analogous to Theorem 6.4 for the two-dimensional $\{11, 13\}$ -generic set is not available.

Problem 6. Describe the set of sterile points. For the Type I family on Σ_7 it is closed and nowhere dense (Proposition 4.17); is the same true for all seven families on Σ_r , and does it have Haar measure zero? The data for the sign fibre and the 17-generic line suggest sterile sets of positive measure *within* these thin fibres (EVIDENCE).

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